

given parameters

this is the same as ϕ

$$h := 12 \text{ m}$$

$$h_t := 1.75 \text{ m}$$

$$t := 0.75 \text{ m}$$

$$f := 0.75 \text{ m}$$

$$\gamma := 20 \frac{\text{kN}}{\text{m}^3}$$

$$\phi := 29 \text{ deg}$$

$$c'_k := 0 \text{ kPa}$$

$$\mu := 0.37$$

$$c'_{ka} := 80 \text{ kPa}$$

$$g_k := 30 \frac{\text{kN}}{\text{m}^2}$$

$$q_k := 24 \frac{\text{kN}}{\text{m}^2}$$

$$l := 0.8 \cdot h = 9.6 \text{ m}$$

$$f_y := 27.5 \frac{\text{kN}}{\text{cm}^2}$$

$$k_1 := 0.5$$

$$k_2 := 30$$

$$n := 0.8$$

$$\tau := 50$$

$$\gamma_{MO} := 1$$

$$\gamma_r := 1.1$$

$$\gamma_R := 1.4$$

$$\gamma_q := 1.35$$

$$\gamma_g := 1.35$$

$$\gamma_{MO} := 1$$

$$\gamma_{R.h} := 1.1$$

1-determination of the critical failure surface

to get the D and the ω we have to make linear interpolation

$$\phi = 29 \text{ deg} \quad \phi_1(\text{smaller}) := 25 \text{ deg} \quad \phi_2(\text{bigger}) := 30 \text{ deg}$$

$$\text{ratio} := \frac{\phi_2(\text{bigger}) - \phi}{\phi_2(\text{bigger}) - \phi_1(\text{smaller})} = \frac{1}{5}$$

$$\omega_1(\text{bigger}) := 43.5 \text{ deg} \quad \omega_2(\text{smaller}) := 38 \text{ deg}$$

$$\omega := \omega_2(\text{smaller}) - \text{ratio} \cdot (\omega_2(\text{smaller}) - \omega_1(\text{bigger})) = 39.1 \text{ deg}$$

$$\omega = 39.1 \text{ deg}$$

$$k_{c1}(\text{bigger}) := 0.26 \quad k_{c2}(\text{smaller}) := 0.21$$

$$k_c := k_{c2}(\text{smaller}) - \text{ratio} \cdot (k_{c2}(\text{smaller}) - k_{c1}(\text{bigger})) = 0.22$$

$$k_c = 0.22$$

$$k_{q1}(\text{bigger}) := 0.70 \quad k_{q2}(\text{smaller}) := 0.60$$

$$k_q := k_{q2}(\text{smaller}) - \text{ratio} \cdot (k_{q2}(\text{smaller}) - k_{q1}(\text{bigger})) = 0.62$$

$$k_q = 0.62$$

$$D_1(\text{bigger}) := 0.43 \cdot h = 5.16 \text{ m} \quad D_2(\text{smaller}) := 0.37 \cdot h = 4.44 \text{ m}$$

$$D := D_2(\text{smaller}) - \text{ratio} \cdot (D_2(\text{smaller}) - D_1(\text{bigger})) = 4.584 \text{ m}$$

$$D = 4.584 \text{ m}$$

based on the information above and from the drawing we determine the following

$$\phi := 29 \text{ deg} \quad \varepsilon := 90 \text{ deg} - (\omega + \phi) = 21.9 \text{ deg} \quad r_A := 13.84 \text{ m} \quad r_B := 20.165 \text{ m}$$

from this point inwards you have to know your ω and then count how many full five degrees it has so you can get the number of

$$\omega_1 := 5 \text{ deg} \quad \omega_2 := 10 \text{ deg} \quad \omega_3 := 15 \text{ deg} \quad \omega_4 := 20 \text{ deg} \quad \omega_5 := 25 \text{ deg} \quad \omega_6 := 30 \text{ deg} \quad \omega_7 := 35 \text{ deg}$$

$$r_1 := r_A \cdot e^{\omega_1 \cdot \tan(\phi)} = 1452.5933 \text{ cm}$$

$$r_2 := r_A \cdot e^{\omega_2 \cdot \tan(\phi)} = 1524.5862 \text{ cm}$$

$$r_3 := r_A \cdot e^{\omega_3 \cdot \tan(\phi)} = 1600.1472 \text{ cm}$$

$$r_4 := r_A \cdot e^{\omega_4 \cdot \tan(\phi)} = 1679.4531 \text{ cm}$$

$$r_5 := r_A \cdot e^{\omega_5 \cdot \tan(\phi)} = 1762.6896 \text{ cm}$$

$$r_6 := r_A \cdot e^{\omega_6 \cdot \tan(\phi)} = 1850.0514 \text{ cm}$$

$$r_7 := r_A \cdot e^{\omega_7 \cdot \tan(\phi)} = 1941.7429 \text{ cm}$$

2-calculation of earth pressure

$$k_{cs} := \frac{1 - (\sin(\phi))^2}{1 + (\sin(\phi))^2} = 0.6194$$

$$H_0 := \frac{k_c}{k_{cs}} \cdot h = 4.2623 \text{ m}$$

$$\sigma_{hAf} := k_q \cdot (g_k + 1.2 \cdot q_k) = 36.456 \text{ kPa}$$

$$\sigma_{hBf} := 1.2 \cdot k_q \cdot g_k = 22.32 \text{ kPa}$$

$$\sigma_{hAH0} := k_c \cdot \gamma \cdot H_0 = 18.7542 \text{ kPa}$$

$$\sigma_{hBH0} := 1.2 \cdot k_c \cdot \gamma \cdot H_0 = 22.5051 \text{ kPa}$$

$$h_{intersection} := 3.31 \text{ m}$$

$$\sigma_{intersection} := 22.464 \text{ kPa}$$

this can be used to calculate the horizontal pressure for the line on the final horizontal graph from the CAD drawing

$$\sigma_{hi}(h_i) := \sigma_{hAf} - \frac{\sigma_{hAf} - \sigma_{intersection}}{h_{intersection}} \cdot h_i \text{ m}$$

e.g.

$$\sigma_{hi}(2.625) = 25.3596 \text{ kPa}$$

this will work until the numbers of strings is in less than

the rest of the spoints or reinforcing strips on the graph will have the same horizontal pressure since if i added 0.75m to 2.625m i will pass my H0
the next strip will be the be the 5th strip which will be

$$\sigma_{h5} := 22.473 \text{ kPa}$$

$$\sigma_{h6} := 22.499 \text{ kPa}$$

$$\sigma_{rest} := \sigma_{hBH0} = 22.5051 \text{ kPa}$$

3-steel reinforcing strip dimensioning

$$N_{ik}(h_i) := \sigma_{hi}(h_i) \cdot t \cdot f$$

e.g.

$$N_{ik}(2.625) = 14.2648 \text{ kN}$$

this will work until the numbers of strings is in less than

after that we can just use this

$$N_{ik5} := \sigma_{h5} \cdot t \cdot f = 12.6411 \text{ kN}$$

$$N_{ik6} := \sigma_{h6} \cdot t \cdot f = 12.6557 \text{ kN}$$

$$N_{hBH0k} := \sigma_{hBH0} \cdot t \cdot f = 12.6591 \text{ kN}$$

$$N_{rest.k} := N_{hBH0k} = 12.6591 \text{ kN}$$

$$\Sigma N_{i.k} := N_{ik}(0.375) + N_{ik}(1.125) + N_{ik}(1.875) + N_{ik}(2.625) + N_{ik5} + N_{ik6} + N_{rest.k} \cdot 10 = 219.647 \text{ kN}$$

here 10 is the number of the strips after the 6th strip until the end

since we design for unfavourable permanent action so

$$N_{id}(h_i) := \gamma_g \cdot N_{ik}(h_i)$$

e.g.

$$N_{id}(1.125) = 24.0725 \text{ kN}$$

this will work until the numbers of strings is in less than

after that we can just use this

$$N_{id5} := \gamma_g \cdot N_{ik5} = 17.0654 \text{ kN}$$

$$N_{id6} := \gamma_g \cdot N_{ik6} = 17.0852 \text{ kN}$$

$$N_{hBH0d} := \gamma_g \cdot N_{hBH0k} = 17.0898 \text{ kN}$$

$$N_{rest.d} := \gamma_g \cdot N_{rest.k} = 17.0898 \text{ kN}$$

now we have to calculate the vertical stresses

$$\sigma_{vi}(h_i) := h_i \cdot \gamma + \sigma_k$$

e.g.

$$\sigma_{vi}(10.125) = 232.5 \text{ kPa}$$

here we have to get the s_i

$$D_1 := 4.59 \text{ m} \quad D_2 := 4.59 \text{ m} \quad D_3 := 4.54 \text{ m} \quad D_4 := 4.45 \text{ m}$$

$$D_5 := 4.32 \text{ m} \quad D_6 := 4.15 \text{ m} \quad D_7 := 3.94 \text{ m} \quad D_8 := 3.695 \text{ m} \quad D_9 := 3.41 \text{ m} \quad D_{10} := 3.085 \text{ m}$$

$$D_{11} := 2.725 \text{ m} \quad D_{12} := 2.33 \text{ m} \quad D_{13} := 1.89 \text{ m} \quad D_{14} := 1.415 \text{ m} \quad D_{15} := 0.89 \text{ m} \quad D_{16} := 0.315 \text{ m}$$

$$s_1 := \gamma_r \cdot \frac{N_{id}(0.375)}{2 \cdot (l - D_1) \cdot \mu \cdot \sigma_{vi}(0.375)} = 0.2 \text{ m}$$

$$s_2 := \gamma_r \cdot \frac{N_{id}(1.125)}{2 \cdot (l - D_2) \cdot \mu \cdot \sigma_{vi}(1.125)} = 0.136 \text{ m}$$

$$s_3 := \gamma_r \cdot \frac{N_{id}(1.875)}{2 \cdot (l - D_3) \cdot \mu \cdot \sigma_{vi}(1.875)} = 0.094 \text{ m}$$

$$s_5 := \gamma_r \cdot \frac{N_{id5}}{2 \cdot (l - D_5) \cdot \mu \cdot \sigma_{vi}(3.375)} = 0.049 \text{ m}$$

$$s_7 := \gamma_r \cdot \frac{N_{rest.d}}{2 \cdot (l - D_7) \cdot \mu \cdot \sigma_{vi}(4.875)} = 0.035 \text{ m}$$

$$s_9 := \gamma_r \cdot \frac{N_{rest.d}}{2 \cdot (l - D_9) \cdot \mu \cdot \sigma_{vi}(6.375)} = 0.026 \text{ m}$$

$$s_{11} := \gamma_r \cdot \frac{N_{rest.d}}{2 \cdot (l - D_{11}) \cdot \mu \cdot \sigma_{vi}(7.875)} = 0.02 \text{ m}$$

$$s_{13} := \gamma_r \cdot \frac{N_{rest.d}}{2 \cdot (l - D_{13}) \cdot \mu \cdot \sigma_{vi}(9.375)} = 0.015 \text{ m}$$

$$s_{15} := \gamma_r \cdot \frac{N_{rest.d}}{2 \cdot (l - D_{15}) \cdot \mu \cdot \sigma_{vi}(10.875)} = 0.012 \text{ m}$$

$$s_4 := \gamma_r \cdot \frac{N_{rest.d}}{2 \cdot (l - 0 \text{ m}) \cdot \mu \cdot \sigma_{vi}(12)} = 0.0098 \text{ m}$$

$$s_4 := \gamma_r \cdot \frac{N_{id}(2.625)}{2 \cdot (l - D_4) \cdot \mu \cdot \sigma_{vi}(2.625)} = 0.067 \text{ m}$$

$$s_6 := \gamma_r \cdot \frac{N_{id6}}{2 \cdot (l - D_6) \cdot \mu \cdot \sigma_{vi}(4.125)} = 0.041 \text{ m}$$

$$s_8 := \gamma_r \cdot \frac{N_{rest.d}}{2 \cdot (l - D_8) \cdot \mu \cdot \sigma_{vi}(5.625)} = 0.03 \text{ m}$$

$$s_{10} := \gamma_r \cdot \frac{N_{rest.d}}{2 \cdot (l - D_{10}) \cdot \mu \cdot \sigma_{vi}(7.125)} = 0.023 \text{ m}$$

$$s_{12} := \gamma_r \cdot \frac{N_{rest.d}}{2 \cdot (l - D_{12}) \cdot \mu \cdot \sigma_{vi}(8.625)} = 0.017 \text{ m}$$

$$s_{14} := \gamma_r \cdot \frac{N_{rest.d}}{2 \cdot (l - D_{14}) \cdot \mu \cdot \sigma_{vi}(10.125)} = 0.013 \text{ m}$$

$$s_{16} := \gamma_r \cdot \frac{N_{rest.d}}{2 \cdot (l - D_{16}) \cdot \mu \cdot \sigma_{vi}(11.625)} = 0.01 \text{ m}$$

$$s_{apl.1} := 20 \text{ cm}$$

$$s_{apl.2} := 15 \text{ cm}$$

$$s_{apl.3} := 10 \text{ cm}$$

$$s_{apl.4} := 8 \text{ cm}$$

$$s_{apl.5} := 6 \text{ cm}$$

$$s_{apl.6} := 6 \text{ cm}$$

$$s_{apl.7} := 4 \text{ cm}$$

$$s_{apl.8} := 4 \text{ cm}$$

$$s_{apl.9} := 4 \text{ cm}$$

$$s_{apl.10} := 4 \text{ cm}$$

$$s_{apl.11} := 4 \text{ cm}$$

$$s_{apl.12} := 4 \text{ cm}$$

$$s_{apl.13} := 4 \text{ cm}$$

$$s_{apl.14} := 4 \text{ cm}$$

$$s_{apl.15} := 4 \text{ cm}$$

$$s_{apl.16} := 4 \text{ cm}$$

$$s_{apl.17} := s_{apl.16}$$

here we have to get the K_1

$$K_1 := \frac{2 \cdot \mu \cdot s_{apl.1} \cdot (l - D_1) \cdot \sigma_{vi}(0.375)}{N_{ik}(0.375)} = 1.42$$

$$K_3 := \frac{2 \cdot \mu \cdot s_{apl.3} \cdot (l - D_3) \cdot \sigma_{vi}(1.875)}{N_{ik}(1.875)} = 1.57$$

$$K_5 := \frac{2 \cdot \mu \cdot s_{apl.5} \cdot (l - D_5) \cdot \sigma_{vi}(3.375)}{N_{ik5}} = 1.81$$

$$K_7 := \frac{2 \cdot \mu \cdot s_{apl.7} \cdot (l - D_7) \cdot \sigma_{vi}(4.875)}{N_{rest.k}} = 1.69$$

$$K_9 := \frac{2 \cdot \mu \cdot s_{apl.9} \cdot (l - D_9) \cdot \sigma_{vi}(6.375)}{N_{rest.k}} = 2.28$$

$$K_2 := \frac{2 \cdot \mu \cdot s_{apl.2} \cdot (l - D_2) \cdot \sigma_{vi}(1.125)}{N_{ik}(1.125)} = 1.64$$

$$K_4 := \frac{2 \cdot \mu \cdot s_{apl.4} \cdot (l - D_4) \cdot \sigma_{vi}(2.625)}{N_{ik}(2.625)} = 1.76$$

$$K_6 := \frac{2 \cdot \mu \cdot s_{apl.6} \cdot (l - D_6) \cdot \sigma_{vi}(4.125)}{N_{ik6}} = 2.15$$

$$K_8 := \frac{2 \cdot \mu \cdot s_{apl.8} \cdot (l - D_8) \cdot \sigma_{vi}(5.625)}{N_{rest.k}} = 1.97$$

$$K_{10} := \frac{2 \cdot \mu \cdot s_{apl.10} \cdot (l - D_{10}) \cdot \sigma_{vi}(7.125)}{N_{rest.k}} = 2.63$$

$$K_{11} := \frac{2 \cdot \mu \cdot s_{apl.11} \cdot (l - D_{11}) \cdot \sigma_{vi}(7.875)}{N_{rest.k}} = 3.01$$

$$K_{12} := \frac{2 \cdot \mu \cdot s_{apl.12} \cdot (l - D_{12}) \cdot \sigma_{vi}(8.625)}{N_{rest.k}} = 3.44$$

$$K_{13} := \frac{2 \cdot \mu \cdot s_{apl.13} \cdot (l - D_{13}) \cdot \sigma_{vi}(9.375)}{N_{rest.k}} = 3.92$$

$$K_{14} := \frac{2 \cdot \mu \cdot s_{apl.14} \cdot (l - D_{14}) \cdot \sigma_{vi}(10.125)}{N_{rest.k}} = 4.45$$

$$K_{15} := \frac{2 \cdot \mu \cdot s_{apl.15} \cdot (l - D_{15}) \cdot \sigma_{vi}(10.875)}{N_{rest.k}} = 5.04$$

$$K_{16} := \frac{2 \cdot \mu \cdot s_{apl.16} \cdot (l - D_{16}) \cdot \sigma_{vi}(11.625)}{N_{rest.k}} = 5.7$$

$$K_{16} := \frac{2 \cdot \mu \cdot 4 \text{ cm} \cdot (l - 0 \text{ m}) \cdot \sigma_{vi}(12)}{N_{rest.k}} = 6.06$$

Sufficient, if $k_1 \geq 1,1 \cdot 1,35 \approx 1,5$

Do it for each strip row!

$$s_{apl.sum} := s_{apl.1} + s_{apl.2} + s_{apl.3} + s_{apl.4} + s_{apl.5} + s_{apl.6} + s_{apl.7} + s_{apl.8} + s_{apl.9} + s_{apl.10} + s_{apl.11} + s_{apl.12} + s_{apl.13} + s_{apl.14} + s_{apl.15} + s_{apl.16}$$

$$l_{sum} := (l - D_1) + (l - D_2) + (l - D_3) + (l - D_4) + (l - D_5) + (l - D_6) + (l - D_7) + (l - D_8) + (l - D_9) + (l - D_{10}) + (l - D_{11}) + (l - D_{12}) + (l - D_{13}) + (l - D_{14}) + (l - D_{15}) + (l - D_{16})$$

effective stress at the lowest row of reinforcing strips

$$\sigma_{vi}(11.625) = 262.5 \text{ kPa}$$

$$N_{ik}(0) = 20.5065 \text{ kN}$$

$$N_{id}(0) = 27.6838 \text{ kN}$$

$$K_n := \frac{2 \cdot \mu \cdot s_{apl.sum} \cdot l_{sum} \cdot \sigma_{vi}(11.625)}{\Sigma N_{i,k}} = 95.8911$$

$$v_0 := \frac{N_{id}(0)}{0.2 \text{ m} \cdot f_y} = 0.5033 \text{ mm}$$

$$v_1 := \frac{N_{id}(0.375)}{s_{apl.1} \cdot f_y} = 0.48 \text{ mm}$$

$$v_2 := \frac{N_{id}(1.125)}{s_{apl.2} \cdot f_y} = 0.58 \text{ mm}$$

$$v_3 := \frac{N_{id}(1.875)}{s_{apl.3} \cdot f_y} = 0.79 \text{ mm}$$

$$v_4 := \frac{N_{id}(2.625)}{s_{apl.4} \cdot f_y} = 0.88 \text{ mm}$$

$$v_5 := \frac{N_{id5}}{s_{apl.5} \cdot f_y} = 1.03 \text{ mm}$$

$$v_6 := \frac{N_{id6}}{s_{apl.6} \cdot f_y} = 1.04 \text{ mm}$$

$$v_7 := \frac{N_{rest.d}}{s_{apl.7} \cdot f_y} = 1.55 \text{ mm}$$

$$v_8 := \frac{N_{rest.d}}{s_{apl.8} \cdot f_y} = 1.55 \text{ mm}$$

$$v_9 := \frac{N_{rest.d}}{s_{apl.9} \cdot f_y} = 1.55 \text{ mm}$$

$$v_{10} := \frac{N_{rest.d}}{s_{apl.10} \cdot f_y} = 1.55 \text{ mm}$$

$$v_{11} := \frac{N_{rest.d}}{s_{apl.11} \cdot f_y} = 1.55 \text{ mm}$$

$$v_{12} := \frac{N_{rest.d}}{s_{apl.12} \cdot f_y} = 1.55 \text{ mm}$$

$$v_{17} := v_{16}$$

$$v_{13} := \frac{N_{rest.d}}{s_{apl.13} \cdot f_y} = 1.55 \text{ mm}$$

$$v_{14} := \frac{N_{rest.d}}{s_{apl.14} \cdot f_y} = 1.55 \text{ mm}$$

$$v_{15} := \frac{N_{rest.d}}{s_{apl.15} \cdot f_y} = 1.55 \text{ mm}$$

$$v_{16} := \frac{N_{rest.d}}{s_{apl.16} \cdot f_y} = 1.55 \text{ mm}$$

$$N_{pl.Rd}(x) := \frac{s_{apl.x} \cdot v_x \cdot f_y}{Y_{M0}}$$

if

$$N_{i,d} \geq N_{pl,Rd}$$

$$N_{pl.Rd0} := \frac{0.2 \text{ m} \cdot 0.5 \text{ mm} \cdot f_y}{Y_{M0}} = 27.5 \text{ kN}$$

$$N_{pl.Rd1} := \frac{s_{apl.1} \cdot v_1 \cdot f_y}{Y_{M0}} = 26.48 \text{ kN}$$

$$N_{pl.Rd2} := \frac{s_{apl.2} \cdot v_2 \cdot f_y}{Y_{M0}} = 24.0725 \text{ kN}$$

$$N_{pl.Rd3} := \frac{s_{apl.3} \cdot v_3 \cdot f_y}{Y_{M0}} = 21.665 \text{ kN}$$

$$N_{pl.Rd4} := \frac{s_{apl.4} \cdot v_4 \cdot f_y}{Y_{M0}} = 19.2575 \text{ kN}$$

$$N_{pl.Rd5} := \frac{s_{apl.5} \cdot v_5 \cdot f_y}{Y_{M0}} = 17.0654 \text{ kN}$$

$$N_{pl.Rd6} := \frac{s_{apl.6} \cdot v_6 \cdot f_y}{Y_{M0}} = 17.0852 \text{ kN}$$

$$N_{pl.Rd7} := \frac{s_{apl.7} \cdot v_7 \cdot f_y}{Y_{M0}} = 17.0898 \text{ kN}$$

$$N_{pl.Rd8} := \frac{s_{apl.8} \cdot v_8 \cdot f_y}{Y_{M0}} = 17.0898 \text{ kN}$$

$$N_{pl.Rd9} := \frac{s_{apl.9} \cdot v_9 \cdot f_y}{Y_{M0}} = 17.0898 \text{ kN}$$

$$N_{pl.Rd10} := \frac{s_{apl.10} \cdot v_{10} \cdot f_y}{Y_{M0}} = 17.0898 \text{ kN}$$

$$N_{pl.Rd11} := \frac{s_{apl.11} \cdot v_{11} \cdot f_y}{Y_{M0}} = 17.0898 \text{ kN}$$

$$N_{pl.Rd12} := \frac{s_{apl.12} \cdot v_{12} \cdot f_y}{Y_{M0}} = 17.0898 \text{ kN}$$

$$N_{pl.Rd13} := \frac{s_{apl.13} \cdot v_{13} \cdot f_y}{Y_{M0}} = 17.0898 \text{ kN}$$

$$N_{pl.Rd14} := \frac{s_{apl.14} \cdot v_{14} \cdot f_y}{Y_{M0}} = 17.0898 \text{ kN}$$

$$N_{pl.Rd15} := \frac{s_{apl.15} \cdot v_{15} \cdot f_y}{Y_{M0}} = 17.0898 \text{ kN}$$

$$N_{pl.Rd16} := \frac{s_{apl.16} \cdot v_{16} \cdot f_y}{Y_{M0}} = 17.0898 \text{ kN}$$

$$N_{pl.Rd0} := \frac{v_{17} \cdot s_{apl.17} \cdot f_y}{Y_{M0}} = 17.0898 \text{ kN}$$

$$N_{pl.Rd16} := \frac{4 \text{ cm} \cdot 1.55 \text{ mm} \cdot f_y}{Y_{M0}} = 17050 \text{ N}$$

from here we can see that they're equal in every strip

$$v_k := k_1 \cdot (\tau - k_2)^n = 5.4928$$

$$v_k := 5.4928 \text{ mm}$$

$$v_{apl.1} := v_1 + v_k = 5.97 \text{ mm}$$

$$v_{apl.2} := v_2 + v_k = 6.08 \text{ mm}$$

$$v_{apl.3} := v_3 + v_k = 6.28 \text{ mm}$$

$$v_{apl.4} := v_4 + v_k = 6.37 \text{ mm}$$

$$v_{apl.5} := v_5 + v_k = 6.53 \text{ mm}$$

$$v_{apl.6} := v_6 + v_k = 6.53 \text{ mm}$$

$$v_{apl.7} := v_7 + v_k = 7.05 \text{ mm}$$

$$v_{apl.8} := v_8 + v_k = 7.05 \text{ mm}$$

$$v_{apl.9} := v_9 + v_k = 7.05 \text{ mm}$$

$$v_{apl.10} := v_{10} + v_k = 7.05 \text{ mm}$$

$$v_{apl.11} := v_{11} + v_k = 7.05 \text{ mm}$$

$$v_{apl.12} := v_{12} + v_k = 7.05 \text{ mm}$$

$$v_{apl.13} := v_{13} + v_k = 7.05 \text{ mm}$$

$$v_{apl.14} := v_{14} + v_k = 7.05 \text{ mm}$$

$$v_{apl.15} := v_{15} + v_k = 7.05 \text{ mm}$$

$$v_{apl.16} := v_{16} + v_k = 7.05 \text{ mm}$$

analysis of sliding

$$\delta := \varphi \quad \beta := 0 \quad \alpha := 0$$

$$K_a := \frac{\cos(\delta) \cdot (\cos(\varphi + \alpha))^2}{(\cos(\alpha))^2 \cdot \left(\sqrt{\cos(\alpha - \delta)} + \sqrt{\frac{\sin(\varphi + \delta) \cdot \sin(\varphi - \beta)}{\cos(\alpha + \beta)}} \right)^2} = 0.2692$$

$$P_{a.g.h.k} := \frac{h^2 \cdot \gamma \cdot K_a}{2} = 387.6854 \frac{\text{kN}}{\text{m}}$$

$$P_{a.q.h.k} := q_k \cdot h \cdot K_a = 77.5371 \frac{\text{kN}}{\text{m}}$$

$$E_d := \gamma_g \cdot P_{a.g.h.k} + \gamma_q \cdot P_{a.q.h.k} = 628.0503 \frac{\text{kN}}{\text{m}}$$

$$R_d := \left(P_{a.g.v.k} + P_{a.q.v.k} + G_k \right) \cdot \frac{\tan(\delta_b)}{\gamma_{R.h}} + \frac{c'_{k.A} \cdot l}{\gamma_{R.h}} + \frac{P_{p.k}}{\gamma_{R.e}}$$

fulfilled, if $E_d \leq R_d$ or $\Lambda \leq 1.0$.

in our case we have to work with cohesive soil surface which is clay

$$R_d := \frac{c'_{ka} \cdot l}{\gamma_{R.h}} = 698.1818 \frac{\text{kN}}{\text{m}}$$

in our case we have to work with granular soil surface

$$G_K := \gamma \cdot h \cdot 1 \text{ m} = 240 \frac{\text{kN}}{\text{m}}$$

$$\tan(\delta_b) := \tan(\delta)$$

$$R_d := \left(G_K \right) \cdot \frac{\tan(\delta_b)}{\gamma_{R.h}} = 120.9402 \frac{\text{kN}}{\text{m}}$$

$$\Lambda := \frac{E_d}{R_d} = 5.1931$$

```

if  $\lambda < 1$ 
  suffecient
else
  notsuffecient

```

retaining wall is not sufficiently secured against sliding.

analysis of bearing capacity

$$R_K := 2 \cdot c'_{ka} \cdot \left(1 + \frac{n}{2}\right) = 411.3274 \text{ kPa}$$

$$R_d := \frac{R_K}{\gamma_R} = 293.8053 \text{ kPa}$$

$$g_d := g_k \cdot \gamma_g$$

$$q_d := q_k \cdot \gamma_q$$

$$V_d := g_d + q_d + \gamma_g \cdot h \cdot \gamma = 396.9 \text{ kPa}$$

fullfilled if

$$R_d \geq E_d$$

not fullfilled

$$h_{max} := \frac{R_d - (g_d + q_d)}{\gamma_g \cdot \gamma} = 8.1817 \text{ m}$$